

WASP-104b

R-2156

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_250207 · created 2026-08-12T23:09:32+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460713.8886 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.044
Transit depth (Rp/Rs)^2	2.27 +/- 1.32 [%]
Orbital Inclination (inc)	85.52 +/- 2.9
Airmass coefficient 1 (a1)	0.03 +/- 0.019
Airmass coefficient 2 (a2)	2.98 +/- 1.71
Scatter in the residuals of the lightcurve fit is	67.79 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	7.5
Transit Duration (day)	0.074 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.044 vs 0.1214 (deviation 0.67σ)
Duration fit vs published	0.074 d vs 0.0742 d (deviation 0.01σ)
Mid-time O–C	15.9 min
Depth SNR	3.4
Residual scatter	67.79 %

Light curve

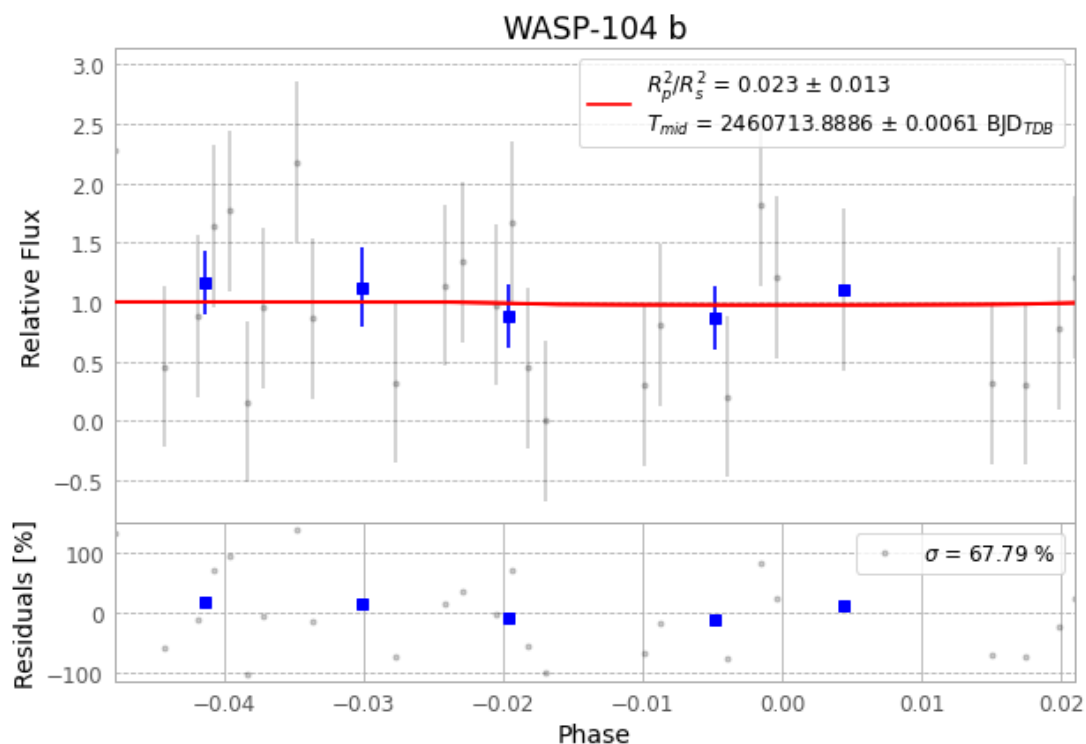


Figure 1 — FinalLightCurve_WASP-104 b_2025-02-07.png (SHA-256 b4068d1959cca16c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [292, 462], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 877d3af026b6306b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

61 FITS frames (40 MB), WASP-104250207070318.FITS → WASP-104250207100315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-250207.log (e76bd8e42a0e...), FinalLightCurve_WASP-104 b_2025-02-07.png (b4068d1959cc...), FinalLightCurve_WASP-104 b_2025-02-07.csv (10afebbcc8585...)

Compute resources

Wall time 179.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:09Z.